**1. Core model: City selection + vehicle deployment + multiple penalties**

$$\min_{y,q,O,U} \quad \underbrace{\sum_c F_c y_c}_{\text{Entry costs}} + \underbrace{\sum_c K_c q_c}_{\text{Vehicle costs}} + \frac{1}{|S|}\sum_s \sum_c \left[ \underbrace{c_c^+ O_c^s}_{\text{Remaining capacity}} + \underbrace{c_c^- U_c^s}_{\text{Foregone contribution margin}} \right].$$

In $c^+, c^-$, $+, -$ distinguish excess and shortage, rather than the signs of costs. All demand, throughput, and costs use **the same evaluation period.** ※ I suppose that setting the evaluation period is needed, and the model determines the city/vehicle deployment to apply during that period.

**2. Constraints**

$$\begin{aligned} 0 \le q_c \le \bar{q}_c y_c \qquad & \forall c, \\ \sum_c q_c \le Q, \qquad & \\ \kappa_c q_c - M_c p_c^s = O_c^s - U_c^s \qquad & \forall c, s, \\ y_c \in \{0,1\}, \quad q_c \in \mathbb{Z}_{\ge 0}, \qquad & \\ O_c^s, U_c^s \ge 0. \qquad & \end{aligned}$$

| Constraint | Meaning |
|---|---|
| $0 \le q_c \le \bar{q}_c y_c$ | Zero vehicles if the city is not entered; up to the city-specific limit if it is entered |
| $\sum_c q_c \le Q$ | Total vehicle limit |
| $\kappa_c q_c - M_c p_c^s = O_c^s - U_c^s$ | The difference between service capacity and robotaxi demand equals unused capacity minus unmet demand. |

**3. Explanations**

**A. Construct two types of demand**

$$D_c^s = M_c p_c^s, \qquad D_c^{N,s} = M_c(1 - p_c^s)$$

| Symbol | Meaning | Explanation |
|---|---|---|
| $M_c$ | Number of **potential ride-hailing usage opportunities** in city $c$ | Robotaxi demand + human-driven taxi demand + other demand |
| $p_c^s$ | Probability of choosing robotaxi for that usage opportunity | Need to be specified with reference service conditions such as fares and waiting times |
| $D_c^s$ | Robotaxi choice demand | Demand **before** applying vehicle shortages |
| $D_c^{N,s}$ | Demand not choosing robotaxis | Human-driven taxis, other modes, giving up the trip, etc. |

**B. Constraints**

1) Once demand $D_c^s$ and service capacity $\kappa_c q_c$ are determined, the following two values are calculated based on the deployment outcome; they are not variables that need to be investigated separately. $(x)^+ = \max\{x, 0\}$. Both values are measured in trips.

$$\begin{aligned} O_c^s &= (\kappa_c q_c - D_c^s)^+ \quad \text{Remaining service capacity,} \\ U_c^s &= (D_c^s - \kappa_c q_c)^+ \quad \text{Unserved potential demand.} \end{aligned}$$

## 4. Notation used in the model

| Symbol | Meaning | Type and unit |
|---|---|---|
| $C$ | Set of potential candidate cities | Input set; cities |
| $S$ | Set of demand scenarios | Input set; scenarios |
| $y_c$ | 1 if city $c$ is entered, else 0 | Decision variable; binary (0 or 1) |
| $q_c$ | Number of vehicles deployed in city $c$ | Decision variable; vehicles (integer) |
| $\kappa_c$ | Number of trips one vehicle can serve during the period | Scale parameter; trips/vehicle over the evaluation period. |
| $Q$ | Total number of vehicles available for deployment | Input parameter; vehicles |
| $\bar{q}_c$ | Upper bound on vehicle deployment in each city | Input parameter; vehicles |
| $F_c$ | Fixed cost of city entry and basic infrastructure | Input parameter; currency over the evaluation period |
| $K_c$ | Per-vehicle preparation, deployment, and holding cost over the evaluation period | Input parameter; currency/vehicle over the evaluation period |
| $c_c^+$ | Additional loss per unit of unused service capacity | Input parameter; currency/trip of unused capacity |
| $c_c^-$ | Foregone contribution margin per unit of unserved potential demand | Input parameter; currency/unserved trip |
| $M_c$ | Number of **potential ride-hailing usage opportunities** in city $c$ | Input parameter; usage opportunities over the evaluation period.<br>data or assumption |
| $p_c^s$ | Probability of choosing our company's robotaxi for that usage opportunity | Scenario input; dimensionless probability in $[0, 1]$ |
| $D_c^s$ | Robotaxi choice demand | Derived input; trips over the evaluation period.<br>$D_c^s = M_c p_c^s$ |
| $D_c^{N,s}$ | Demand not choosing robotaxis (human-driven taxis, other modes, giving up the trip, etc.) | Derived input; trips over the evaluation period.<br>$D_c^{N,s} = M_c(1 - p_c^s)$ |
| $O_c^s$ | Unused service capacity | Outcome variable; trips over the evaluation period.<br>$O_c^s = (\kappa_c q_c - D_c^s)^+$ |
| $U_c^s$ | Unserved robotaxi demand | Outcome variable; trips over the evaluation period.<br>$U_c^s = (D_c^s - \kappa_c q_c)^+$ |

※ Demand-related variables consist of the inputs $p_c^s, M_c$ and outcome variables calculated from scenario demand and deployment in the model. For the baseline model, $M_c$ is fixed. ※ Outcome variables are not quantities to be predicted by collecting data for each one separately. They are outcomes calculated from demand and deployment.

[*Moved from* ***baseline(main)*** *file to* ***candidates_(M,p)*** *file in* ***driving factors/Candidates*** *folder.*]

## 5. Driving factors

### A. Driving factors of $M$

The papers below mainly explain **observed ride-hailing usage**, so they do not include the 'unrealized demand' included in the definition of $M$, making direct estimation impossible. However, I suppose the demand patterns provided by these papers lead to constructing 'potential travel opportunities $M$' through modeling.

| Variable | Supporting paper | Scope established in the paper | Measurement data |
|---|---|---|---|
| **Population and population density** | Elkhouly and Alhadidi, *Ride-sharing Determinants* (paper); Zhang et al., *Examining spatial heterogeneity⋯* (paper) | Relationship of population and local density with conventional ride-hailing usage. Density effects may vary by area | ACS |
| **Employment and economic activity** | Elkhouly and Alhadidi (paper) | Uses the economically active population and employment environment to explain usage | LEHD/LODES, ACS |
| **Income** | Elkhouly and Alhadidi (paper) | Analyzes the relationship between income and usage in Chicago | ACS B19013 |
| **Share of car-free households** | Elkhouly and Alhadidi (paper) | Analyzes the relationship between households without cars and usage | ACS B08201 |
| **Land-use mix, accessibility, and commercial activity** | Zhang et al., *Examining spatial heterogeneity⋯* (paper) | Nonlinear and spatial differences in the urban environment and ridesourcing usage | EPA SLD |
| **Restaurant and leisure-facility density** | Loa, Lee, and Circella, *Just another ride-hailing?* (paper) | Relationship with **robotaxi trip counts in SF**. Does not separate the effects of $M$ and $p$ | Census CBP, OpenStreetMap |
| **Public transport and walking environment** | Zhang et al. (paper); Cats et al., *Beyond the dichotomy⋯* (paper) | Analyzes the local environment and travel-time competitiveness and substitutability with public transport. Not a uniform increase/decrease effect | EPA SLD, GTFS |
| **Airport-related market size** | Zhang et al. (paper) | Analyzes airport travel as a separate spatial category. **Does not estimate the direct effect of airport passenger counts** | FAA passenger data. Used as a proxy for market size |
| **Time, day of week, and season** | Elkhouly and Alhadidi (paper) | Analyzes differences in usage over time. Specific variables such as holidays are selected through separate design | Dates and times in trip data |
| **Weather** | Additional: Liu, Jiang, and Chen, *Quantifying the impact of weather on ride-hailing ridership: Evidence from Haikou, China* (2021, paper) | Analyzes the relationship between weather and ride-hailing usage. Effects in U.S. cities require revalidation | NOAA CDO |

### B. Driving factors of $p$

| Variable | Supporting paper | Scope of application | Required measurement data |
|---|---|---|---|
| **Relative fare** | Additional: Yin and Cherchi, *Willingness to pay for automated taxis*⋯ (2024; online 2022, paper) | Choice experiment between conventional taxis and automated taxis | Offered fares for both + choice outcomes |
| **Relative waiting time** | Yin and Cherchi (paper) | Analyzes choices under different service conditions | Offered waiting times for both + choice outcomes |
| **In-vehicle travel time** | Yin and Cherchi (paper) | Analyzes choice trade-offs between fares and time | Offered travel times for both |
| **Trust, price value, and usage habits** | Additional: Wei et al., *Using the Extended Unified Theory*⋯ (2024, paper) | Study of robotaxi acceptance intention and behavior. **Survey scales cannot be interpreted directly as choice probabilities** | Individual attitude and habit surveys |
| **Age and robotaxi-related knowledge** | Yin and Cherchi (paper) | Choice differences by individual characteristics. Fixing the sign, such as "younger people always prefer it, is inappropriate | Link individual characteristics to choice data |
| **User reviews and in-vehicle support features** | Yin and Cherchi (paper) | Analyzes customer ratings and features such as destination changes and communication with an operator as choice attributes | Service attributes in the choice experiment |
| **Experience using robotaxis** | Additional: Yin, Cherchi, and Nettle, *Living in a Ci ty Where Automated Taxis are Operating and Using Them*⋯ (2023, paper) | Analyzes preference differences by whether service operates in the residential area and by individual usage experience | Individual usage experience |
| **Competitiveness of alternatives such as public transport** | Cats et al., *Beyond the dichotomy*⋯ (paper) | Evidence from **conventional ride-hailing**. Not a robotaxi-specific preference effect | Availability, fares, and times of alternatives |